

IDS 570: TEXT AS DATA | TERM PROJECT (OPTION A) REPORT

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1. Introduction

In this project, I used methods of computational text analysis to examine the semantic use of the concept of “money” across a corpus of 571 documents from the Early Modern era. To explore this concept, three main methods were used: Named Entity Recognition; BERT-based clustering, and supervised machine learning.

This project is accompanied by a GitHub folder (available [here](#)). It includes a Jupyter notebook and a folder containing the tables and visualisations generated in this report.

Disclosure on AI: I used ChatGPT to debug and optimize the code, including to extract the sentence windows around the concept of ‘money’ (and ‘moneys/monies’) in the corpus, improving the implementation of sentence tokenization using NLTK, debugging the BERT-clustering and the supervised learning processes, and reducing the time taken to process the corpus. All analytical decisions, methodological choices, and interpretations of the results are my own.

2. Literature review

The mass digitization of historical texts since the turn of the millennium, combined with advancements in digital technology, has made it possible to use computational text analysis methods to uncover patterns in the use of language, terminology and concepts (and shifts or trends in their meaning) across documents. A considerable body of work has been built up on the validity and usefulness of different methods in relation to historical documents, including those employed in this project.

First, several works discuss the use of methods, such as Named Entity Recognition, to identify specific proper names (like institutions, people, or places) as concepts and then map relationships between two or more concepts to uncover broader patterns. Here, Ehrmann et al. (2023) provide a useful account; they discuss the main approaches used to apply Named Entity Recognition (NER) on historical documents and provide an overview of how different challenges - from historiographical complexities to optical character recognition errors - can be addressed. More recent works, such as Woodrum et al. (2025), demonstrate how to leverage recent technological advancements around large language models (LLMs) to conduct NER efficiently; they discuss approaches to leveraging ChatGPT efficiently to identify groups in tagging ancient Latin tasks (Woodrum et al., 2025). While this is not explored in this project, it points to a future direction to consider: using a LLM to identify concepts related to ‘money’ in the corpus.

Second, another stream in the literature is to use transformer-based language models, such as BERT (Devlin et al., 2019), to generate contextualized embeddings for a selection of words, and then analyze them to uncover how words cluster in different settings. Here, adaptation to historical texts often requires pre-training on the historical variant of the language in order to improve model performance. For instance, Manjavacas and Fonteyn (2021) provide a useful account of their development of ‘MacBERh’, a transformer-based language model that is pre-trained on historical English, and how that outperforms models pre-trained on present-day language. While the present analysis does not engage in pre-training and relies on an ‘off-the-shelf’ model due to time constraints, this paper points to the importance of carefully constructing the pipeline through which natural language processing (NLP) methods are applied to historical texts. For now, this paper will primarily focus on using BERT-based clustering to determine the different ways in which ‘money’ and its variants are used in the corpus.

Third, this paper also seeks to use supervised machine learning to classify different instances of the term ‘money’ (and its variants) into their predominant clusters. Two useful accounts are provided by Barberá et al. (2021) and (Sorbi, Moccozet & Marchand-Maillet, 2026). They discuss how machine learning techniques can be effectively adapted to historical texts with weak supervision over a small set of entity labels. These papers provide the basis for the weak supervision approach employed in this project.

3. Data & setup

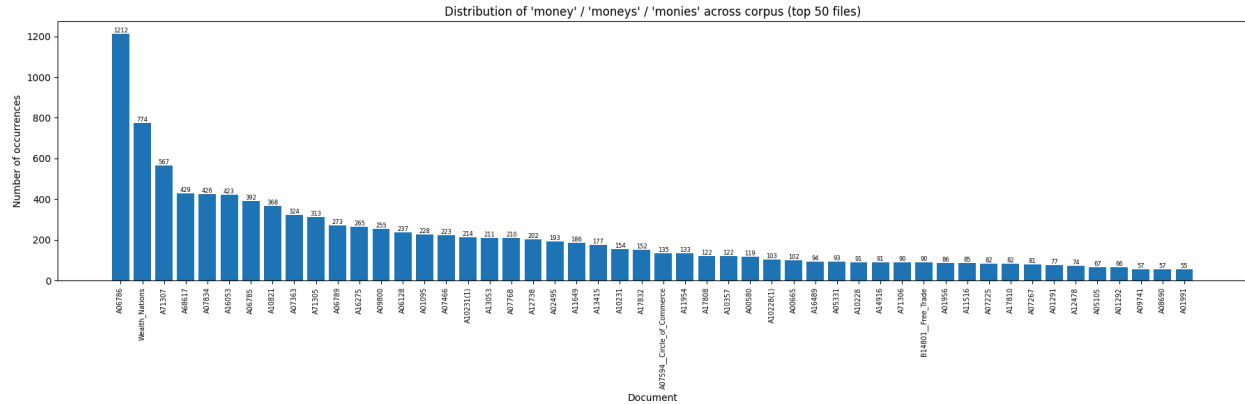
I selected the concept of ‘money’ for this analysis, as this is a term that appears frequently in Early Modern texts, to refer to a medium of exchange, a unit of account, or, more broadly, an indicator of wealth or commercial activity. This concept has historically served several functions - such as recording trade balances, prices, interest rates - and has even (when associated with ‘debt’) acquired moral significance (Graeber, 2011). Consequently, it is central to debates about trade and policy during the era in question.

Note: for this analysis, I created a case-insensitive regular expression in the search for matches across the corpus to capture the different spellings of money and its plural term (specifically, ‘money’, ‘mony’, ‘moneys’, and ‘monies’). The goal was to comprehensively retrieve instances of the concept, including its possible different meanings across contexts. Accordingly, the unit of analysis was a ± 2 sentence window around each occurrence of that regular expression.

3.1. Initial patterns observed

The word ‘money’ (and its variants in scope) are mentioned in 440 out of the 571 documents; I identified a total of 15592 occurrences across the corpus. Looking at the top 50 documents (as graphed below), I noted that a small share of the documents did account for a bulk of these occurrences. The skewed distribution suggests a degree of clustering in the corpus. For instance, there were 1212 occurrences in ‘*The Ancient Law Merchant*’ (1622) by Gerard Malynes’ and 774 occurrences in ‘*The Wealth of Nations*’ (1776) by Adam Smith. Further, after the first ~10 texts, counts drop to below 200 per text and flatten out - which could reflect the fact

that ‘money’ is a common term in general. This pattern suggests that the term ‘money’ tends to appear most frequently in texts concerning international trade rules (Malynes), economics, or discourses on wealth (Smith). This also suggests that there is value in mapping relationships between this term and other concepts, such as institutions, places, or people.

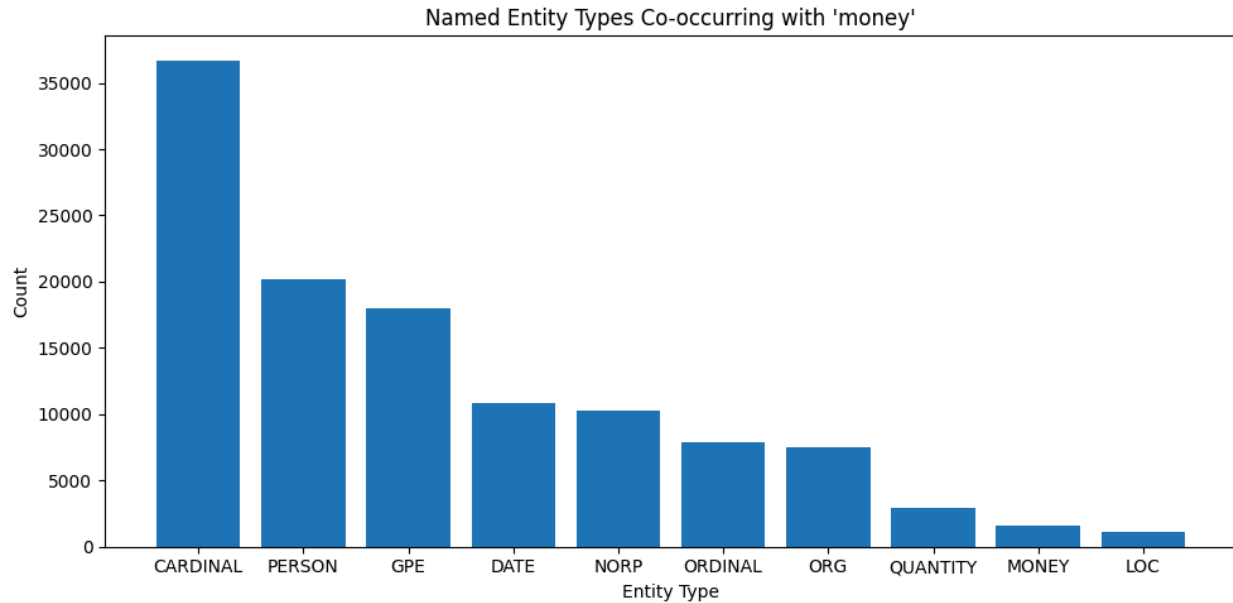


4. Approach 1 – Named Entity Recognition using spaCy

I ran the basic en_core_web_sm on the 15592 sentences with the word “money” (or its variants) to discuss how it is discussed in the corpus. The resulting entity frequency table is as follows:

Entity Type	Count
CARDINAL	36716
PERSON	20197
GPE	17987
DATE	10864
NORP	10297
ORDINAL	7889
ORG	7547
QUANTITY	2893
MONEY	1613
LOC	1144
TIME	1105
LANGUAGE	927
PRODUCT	220
FAC	118

LAW	42
WORK_OF_ART	21
EVENT	20
PERCENT	15



Evidently, the highest frequency count goes to the 'CARDINAL' type, which refers to numbers or numerals that do not fall under other similar quantitative concepts (e.g., ORDINAL, QUANTITY, or MONEY). None of these are very useful, as these merely indicate that the term 'money' is close to quantitative concepts - which makes sense, considering the principal functions of the concepts. As we see below, these are also frequently instances of mislabelling of (what are likely) page numbers or paragraph references by the model. I focused on the next highest set of frequency counts, which have gone to 'PERSON', 'GPE/LOC' (places), and 'ORG' entities. As these are more distinctive entities, associations between them and the term 'money' may be more insightful. Further, I also combine 'NORP' (nationalities) with 'GPE/LOC', as we can consider the two to be interrelated; jointly, this is a reference to a specific distinctive geographical grouping of some sort.

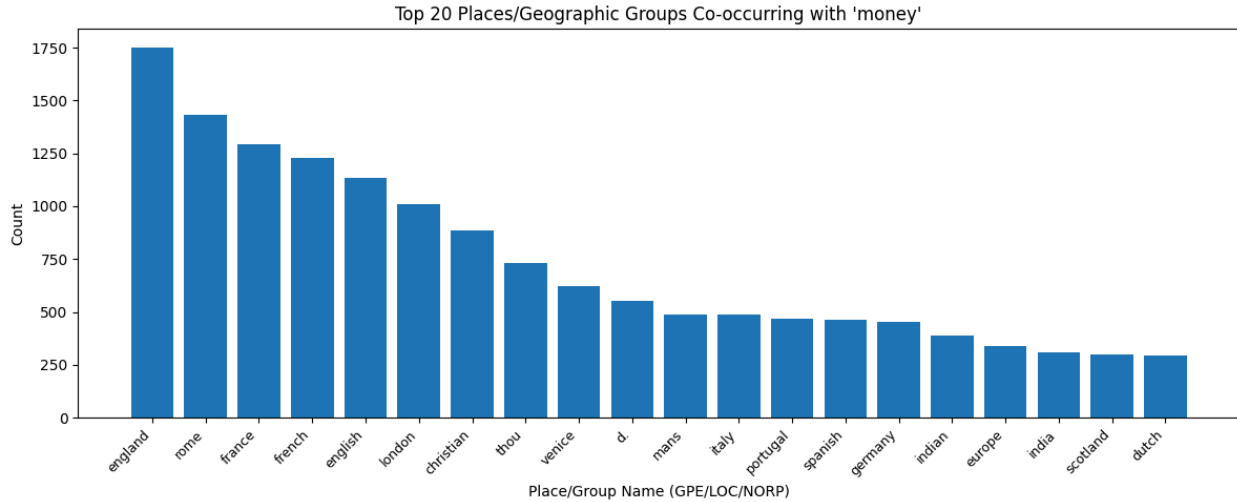
4.1. Interpretations

I note below a summary table of the top 10 entities of each type in focus.

type	entity	count
PLACE/GROUP	england	1751

PLACE/GROUP	rome	1432
PLACE/GROUP	france	1294
PLACE/GROUP	french	1230
PLACE/GROUP	english	1134
PLACE/GROUP	london	1011
PLACE/GROUP	christian	885
PLACE/GROUP	thou	729
PLACE/GROUP	venice	620
PLACE/GROUP	d.	554
PERSON	knowne	301
PERSON	hee	294
PERSON	lib	193
PERSON	hundreth pounds	149
PERSON	philip	135
PERSON	al	124
PERSON	li	123
PERSON	hundreth	123
PERSON	peter	118
PERSON	don	107
ORG	bee	1726
ORG	senate	345
ORG	germanie	158
ORG	shee	114
ORG	neuerthelesse	56
ORG	^e	52
ORG	apollo	42
ORG	mensura	39
ORG	deerer	34
ORG	knowne	32

This clearly shows that the term 'money' does appear to be associated with some distinct places and groups, such as the 'french', 'english', 'london' and 'venice'. The following visual provides a summary of the top 20 places associated with 'money'. However, the other instances of co-occurrence with 'ORG' or 'PERSON' are less useful, largely because the model appears to have significantly mislabelled entities. However, some institutions (e.g., 'the senate' or 'apollo' do appear to be related to money).



4.2. Examples of co-occurrence

I provide below three examples of co-occurrence that support the above conclusion.

Doc ID: A02325(1).txt (shows labelling of CARDINAL)

Window: 7. that shall he also reap not grape of thorn matth. 7. 16 nor fig of thistle can man expect and now though plant of grace like fruitful tree be chief to be wish for in all plantation yet no doubt may civil man be send in case of want as nature flower for ornament i wish therefore that care be have

Entities: 7 [CARDINAL] | 7 [CARDINAL] | 16 [CARDINAL] | fig [PERSON] | the day [DATE] | summer [DATE] | twelve hour [TIME] | twenty four hour [TIME] | day [DATE] | hundred [CARDINAL] | more than one [CARDINAL] | a month [DATE] | twenty penny [MONEY]

Doc ID: A02325(1).txt (shows labelling of CARDINAL)

Window: 7. that shall he also reap not grape of thorn matth. 7. 16 nor fig of thistle can man expect and now though plant of grace like fruitful tree be chief to be wish for in all plantation yet no doubt may civil man be send in case of want as nature flower for ornament i wish therefore that care be have

Entities: 7 [CARDINAL] | 7 [CARDINAL] | 16 [CARDINAL] | fig [PERSON] | the day [DATE] | summer [DATE] | twelve hour [TIME] | twenty four hour [TIME] | day [DATE] | hundred [CARDINAL] | more than one [CARDINAL] | a month [DATE] | twenty penny [MONEY]

Doc ID: A00503.txt (shows relationship with a place)

Window: which is but for wise men to relatum mihi a side digni[^]. feed on: both were big-lookt souldiers: our happy age did bring them: the lofty low-countrys did breed them: amongst the most warlike troupes of europe. the one set to dice, and proved unlucky; his mony being out, his mettle got up, and from
Entities: europe [LOC]

Doc ID: A14614.txt (shows relationship with a group)

Window: rom. 2. 1. but that you may a little be ter consider the weaknesse of this discourse, if the testimony of our selues and our contraries were sufficient and certaine to make tru h, and euer more safe and secure to follow that side which hath that testimony, it had beene better to haue become a iewish
Entities: 2 [CARDINAL] | christian [NORP] | rea|son [PERSON] | pagan [GPE] | christian [NORP]

Doc ID: A14614.txt (shows relationship between money and an organization)

Window: he was not beleueed; so the end there|of was miserable, and drew after it the ruine of great and illu|strious houses of germanie. as for the warre in germanie it began not till after luthers death, neither was it a rebel|lion of the protestants; the truth is they stood for their liues. the emperour
Entities: germanie [ORG] | first [ORDINAL] | eighth [ORDINAL]

Doc ID: A11368.txt (shows relationship between ‘money’ and a person (god))

Window: god; it chanced at last that their pouerty was soe great, as they had nothing to sustaine their life, nor defend them from the extremity of pouerty; at which the mother exceedingly afflicted, had recourse one day vnto the church, where before an image of the b. virgin deuoutly kneeling down, she wit
Entities: god [PERSON] | one day [DATE] | two [CARDINAL] | al humane [PERSON] | fayleth mee [PERSON] | yong [GPE] | a hundred [CARDINAL]

These examples reinforce the different types of usage with which money is associated: (1) with places/groups and (2) with quantitative concepts. The random examples also provide some interesting associations between money and ‘god’. While this does not register a high frequency count and is thus not investigated further, a close reading may reveal an association between ‘money’ and praying to the christian God to alleviate poverty, an interesting strand to explore further.

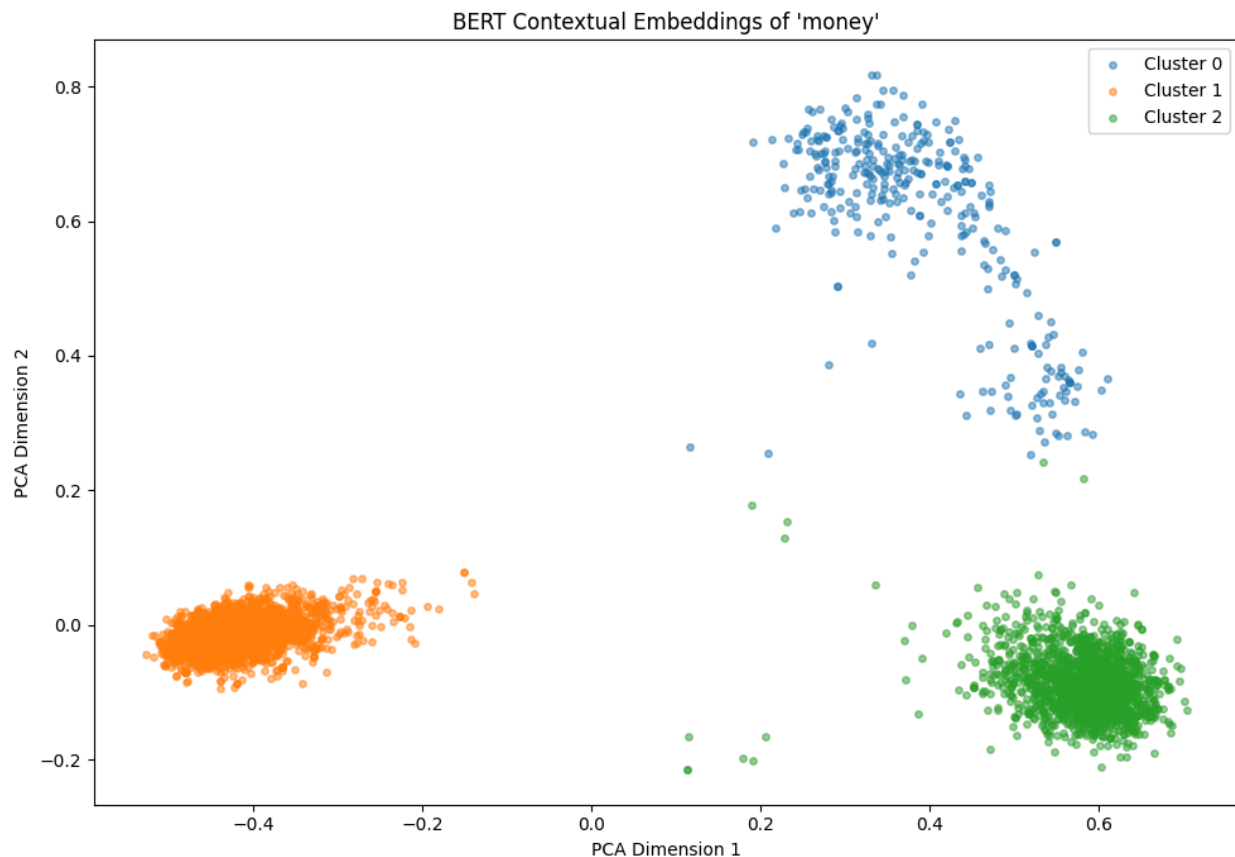
4.3. Limitations of NER in this corpus

I note several limitations. First, the word “money” itself appears as a NER entity; this self-referential tagging may cause us to lose out on capturing more nuanced meanings. Second, as this spaCy model was trained on web text and is not pre-trained on the historical English from the era in question, it may also lack accuracy. This is evidenced by the fact that the model

did not perform well on categorizing and tagging different quantitative concepts - for instance, “hundred pounds” was categorized as a “PERSON”, while “thou” was classified as a “PLACE”. There are also evidently OCR errors distorting frequency counts, as indicated by “li” or “d.” getting high counts.

5. Approach 2 - Contextual meaning with BERT

Next, I used BERT-clustering to analyze how the meaning of “money” may vary across the corpus. I focused on a random sample of 5000 sentences from the full set to make the analysis more manageable. Next, I grouped the embeddings generated into three clusters using KMeans clustering ($k=3$). This is stemming from the analysis from Approach 1, that seemed to suggest that ‘money’ is associated with three distinct groupings - cardinal values, places/groups, and specific persons (including ‘god’). The results were visualized using Principal Component Analysis to reduce the 768-dimensional embeddings to two dimensions. As the graph below shows, there are three clear clusters, which are fairly spread out.



Using representative examples, it appears that we can see three ‘functions’ of money emerge: (1) money as a medium of exchange between people (2) money as a form of consideration and (3) money as a unit of value of some sort that is linked to a specific place or group (or institution). Here, on the third cluster, I note that a unit of value has to be assigned by an

institution. The value of money comes from the institutional backing as pieces of paper or coin have no intrinsic value independent of an institutional guarantee. Accordingly, examples are:

Cluster 0: Doc ID: A07594__Circle_of_Commerce.txt (money: exchange)

to the 3. §. 26. i reply, that although there wanted not takers of mony by exchange for hampburgh, when the exchange risse from 28 shillings to 35 shillings, yet it doth not follow, that therefore there would be takers at his par of exchange: for it is a plaine dispar, a different case. for those that then tooke mony for hampburgh, the takers gained and the deliuerers lost: because the exchange risse faster at hampburgh, by reason of the raising of the monies there, then it did at London. which if m

Cluster 1: Doc ID: A13053.txt (money: consideration)

concerning the indenture of covenants, &c. w hereas (of late) differences have arisen betweene the right honorable thomas I. wentworth, the cause of the now compounding lord of the mannors of stepney and hackney, and his lordships copyhold tenants of the said mannors, for and concerning some of the customes, benefits, and priviledges of the said tenants: it hath now pleased the said thomas lord wentworth, in consideration of three thousand pounds, the consideration given for it. of law fu

Cluster 2: Doc ID: Wealth_Nations.txt (money: institutionally set unit of value - here, the 'london' market)

but though, in establishing perpetual rents, or even in letting very long leases, it may be of use to distinguish between real and nominal price; it is of none in buying and selling, the more common and ordinary transactions of human life. at the same time and place, the real and the nominal price of all commodities are exactly in proportion to one another. the more or less money you get for any commodity, in the london market, for example, the more or less labor it will at that time and place e...

However, as the examples show, due to noise and mislabelling in the dataset, the clusters are not necessarily as neat in content, and can be difficult to interpret consistently. This is also evidenced by the embedding counts: (consideration = 2834; unit_value = 1836; medium_exchange = 330), which suggests that we can consider simplifying to two broad categories of 'consideration' (merging consideration and medium of exchange) and 'institutionally set unit of value' for the next approach.

6. Approach 3 - Supervised Classification (Logistic Regression)

For this section, I rely on a simplified weak labeling scheme that groups the meaning of 'money' into 2 categories: 'consideration' and (institutionally set unit of) 'value'. Next, I train a logistic regression model using this approach on a subset of 25 examples and then apply it to the

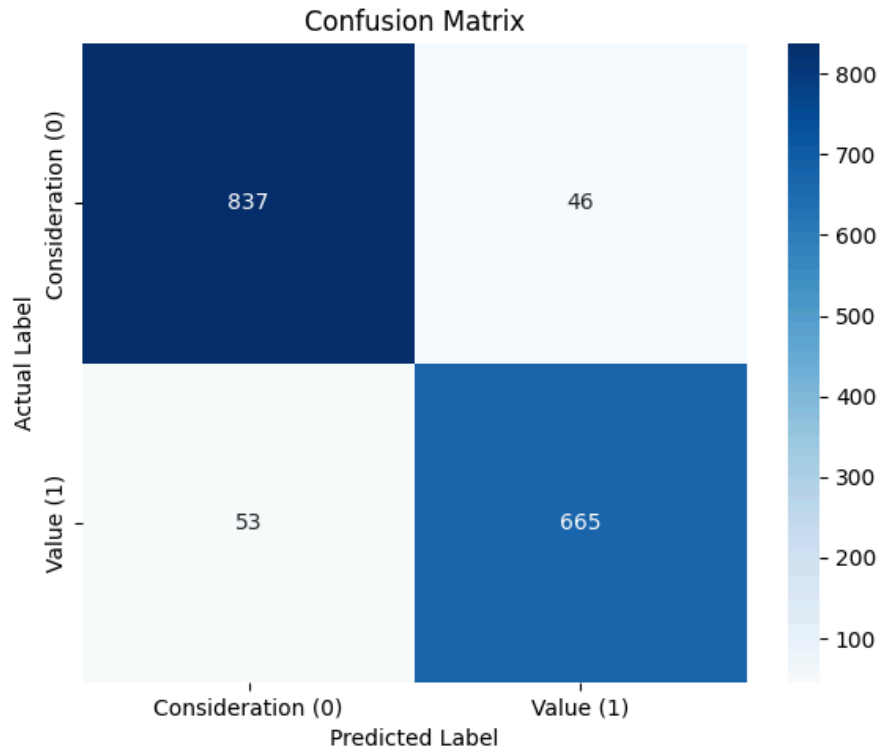
remaining data. The model was trained on 6404 sentences and tested on 1601 sentences. The results are interpreted as follows.

6.1. How well does the model distinguish meanings?

The model demonstrated strong performance on the test, achieving an overall precision of 0.94 on both classes, with slightly higher precision and F-1 scores for the Consideration class (0.95, 0.94) than for the Value class (0.93, 0.93).

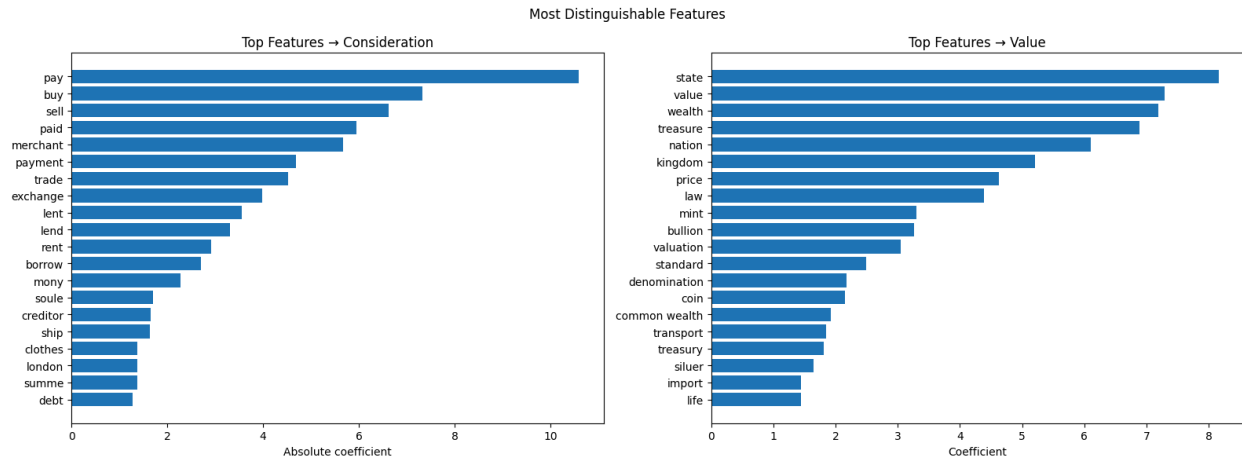
	precision	recall	f1-score	support
Consideration	0.9404494382022470	0.9479048697621740	0.9441624365482230	883.0
Value	0.9353023909985940	0.9261838440111420	0.9307207837648710	718.0
accuracy	0.9381636477201750	0.9381636477201750	0.9381636477201750	0.9381636477201750
macro avg	0.9378759146004200	0.9370443568866580	0.937441610156547	1601.0
weighted avg	0.9381411434538260	0.9381636477201750	0.9381342624704930	1601.0

This is also supported by the confusion matrix, which shows the errors are fairly balanced and not skewed in favor of either class.



6.2. What features seem to drive classification?

Evidently, features pushing sentences towards ‘consideration’ include those with terms referencing exchange or consideration, like pay, buy, sell, etc. Similarly, those pushing sentences towards ‘value’ include terms like ‘state’ and ‘wealth’ and ‘treasure’. This supports the general hypothesis of how the two board clusters have been identified for this analysis.



6.3. Where does the model fail?

The model does not perform well when the meanings of money are not cleanly separated. There are sentences where both functions of money may be at play, leading to misclassifications, as the model cannot accurately parse the overall passage. Another clear instance is the pushing of sentences containing the term ‘value’ towards the ‘Value’ set, which suggests the model may be strongly influenced by keyword signals; this may be because the labels were not generated using fully manual annotation, which means that ambiguous examples are difficult to classify consistently. As the confusion matrix indicates, the errors are roughly balanced, which suggests that the categories may themselves overlap due to the dual meanings of some keywords.

7. Synthesis

What this analysis indicates is that ‘money’ is not a single stable concept; it moves between its different functions. As the BERT clustering indicated, there are three primary clusters of meanings assigned to the term – medium of exchange, consideration, and unit of value. The NER results do correlate to an extent, showing that, at one hand, references to money are tied to quantitative values (supporting the first two clusters), but also to places, groups, and institutions (which supports the last cluster). This is why the classification model - which operationalizes a simplified distinction between ‘Consideration’ and ‘Value’ - performs well and is driven by lexical patterns.

However, we also see the overlap between these categories from the confusion matrix; there are also more ambiguous relationships (such as between ‘money’, ‘poor’, and ‘god’) to consider. Accordingly, the central claim from this analysis is that the meanings of money constitute overlapping and independent functions; it must be constructed contextually for its effective meaning to emerge from a given passage. In fact, I would claim that a term like ‘money’ may only be accurately understood by a ‘close reading’ approach, as quantitative approaches may miss nuance from ambiguous and overlapping meanings.

8. References (APA 7)

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